

Solutions for Problem Set 2

Mathematics for Social Scientists

EXERCISE 1. (*Essential Mathematics for Economic Analysis (5th edition)*, Knut Sydsaeter and Peter Hammond, Section 1.2, Question 2)

Determine which of the following formulas are true. If any formula is false, find a counter example to demonstrate this, using a Venn diagram if you find it helpful.

a. $A \subseteq B \iff A \cup B = B$

True

If $A \subseteq B$, then $A \cup B = B$.

If $A \cup B = B$, then $x \in A \Rightarrow x \in A \cup B \Rightarrow x \in B \Rightarrow A \subseteq B$.

Then, we have $A \subseteq B \Rightarrow A \cup B = B$ and $A \cup B = B \Rightarrow A \subseteq B$, that is,

$$A \subseteq B \iff A \cup B = B^1$$

b. $A \subseteq B \iff A \cap B = A$

True

c. $A \cap B = A \cap C \Rightarrow B = C$

False

d. $A \cup B = A \cup C \Rightarrow B = C$

False

e. $A = B \iff (x \in A \iff x \in B)$

True

EXERCISE 2. (*Essential mathematics for Economic Analysis (5th edition)*, Knut Sydsaeter and Peter Hammond, Section 5.3, Question 7)

If f is the function that tells you how many kilograms of carrots you can buy for a specified amount of money, then what does f^{-1} tell you?

f^{-1} tells us how much carrots cost.

More precisely, f^{-1} tells us how much money it would cost to buy a specified amount of carrots.

EXERCISE 3. (Wherever they exist) find the inverse of the following functions:

a. $y = 1/x^2$

$$f^{-1}(x) = \frac{1}{\sqrt{x}}$$

¹I wrote full solution to 1a. The others, I just provide the answer.

b. $y = \sqrt{\sqrt{x} - 5}$
 $f^{-1}(x) = x^4 + 10x^2 + 25$

c. $y = 3^x$
 $f^{-1}(x) = \log_3(x)$

d. $y = x^2 + 2x + 1$
 $f^{-1} = \sqrt{x} - 1$

EXERCISE 4. Are the following functions injective/surjective/bijective?

(i) $f_1 : \{1, 2, 3\} \rightarrow \{1, 2, 3, 4, 5\}$ with $f_1(1) = 2$, $f_1(2) = 1$ and $f_1(3) = 5$
Injective

(ii) $f_2 : \mathbb{R} \rightarrow \mathbb{R} : x \rightarrow x^2$
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(iii) $f_3 : \mathbb{R} \rightarrow \mathbb{R}_0^+ : x \rightarrow x^2$
Surjective

(iv) $f_4 : \mathbb{Z} \rightarrow \mathbb{Z} : n \rightarrow n + 1$
Bijjective

(v) $f_5 : \mathbb{R} \rightarrow \mathbb{R} : x \rightarrow x^3$
Bijjective

(vi) $f_6 : \mathbb{R}_0^+ \rightarrow \mathbb{R} : x \rightarrow x^3$
Injective

(vii) $f_7 : (0, \infty) \rightarrow (0, \infty) : x \rightarrow \frac{1}{x}$
Bijjective

Notation: $\mathbb{R}_0^+ := \{x \in \mathbb{R} : x \geq 0\}$.

EXERCISE 5. Find the following limits (**without** using the L'Hospital's Rule):

Here is an example:

$$\lim_{x \rightarrow \infty} \frac{2x^3 + 3}{x^3 - 5} = \lim_{x \rightarrow \infty} \frac{2 + \frac{3}{x^3}}{1 - \frac{5}{x^3}} = \frac{\lim_{x \rightarrow \infty} 2 + \frac{3}{x^3}}{\lim_{x \rightarrow \infty} 1 - \frac{5}{x^3}} = \frac{2}{1} = 2$$

(i) $\lim_{x \rightarrow 4} [x^2 - 3x + 4]$

(ii) $\lim_{x \rightarrow 4} \frac{x^2}{3x-5}$

(iii) Given $\lim_{y \rightarrow 0} f(y) = -2$ and $\lim_{y \rightarrow 0} g(y) = 2$, find $\lim_{y \rightarrow 0} [f(y) - 3f(y)g(y)]$.

(iv) $\lim_{x \rightarrow \infty} \frac{11x^2}{x^2+3}$

(v) $\lim_{x \rightarrow \infty} \frac{3x-12}{x+1}$

(vi) $\lim_{x \rightarrow \infty} \frac{2x-3}{2x+5}$

(i) 8

(ii) $\frac{16}{7}$

(iii) 10

(iv) 11

(v) 3

(vi) 1

EXERCISE 6. Find the following limits **without** using L'Hôpital's Rule.

$$(i) \lim_{t \rightarrow -2} \frac{t^3 + 8}{t + 2} = \lim_{t \rightarrow -2} \frac{(t+2)(t^2 - 2t + 4)}{t + 2} = \lim_{t \rightarrow -2} (t^2 - 2t + 4) = 12.$$

$$(ii) \lim_{t \rightarrow -1} \frac{t^3 + t^2 - 5t + 3}{t^2 - 3t + 2} = \frac{4}{3}.$$

$$(iii) \lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4} = \lim_{x \rightarrow 4} \frac{(x-4)(x+4)}{x - 4} = 8.$$

$$(iv) \lim_{x \rightarrow 2} \frac{x^2 - 4x + 4}{x^2 + x - 6} = \lim_{x \rightarrow 2} \frac{(x-2)^2}{(x+3)(x-2)} = \lim_{x \rightarrow 2} \frac{x-2}{x+3} = 0.$$

$$(v) \lim_{t \rightarrow 1} \frac{t^3 + t^2 - 5t + 3}{t^3 - 3t + 2} = \lim_{t \rightarrow 1} \frac{(t-1)^2(t+3)}{(t-1)^2(t+2)} = \frac{4}{3}.$$

$$(vi) \lim_{x \rightarrow 3^+} \frac{x}{x-3} = +\infty.$$

$$(vii) \lim_{x \rightarrow 1^+} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow 1^+} \frac{(x-1)(x^3 + x^2 + x + 1)}{x - 1} = \lim_{x \rightarrow 1^+} (x^3 + x^2 + x + 1) = 4.$$

$$(viii) \lim_{x \rightarrow 4^-} \frac{3-x}{x^2 - 2x - 8} = \lim_{x \rightarrow 4^-} \frac{3-x}{(x-4)(x+2)} = +\infty, \quad \lim_{x \rightarrow 4^+} \frac{3-x}{x^2 - 2x - 8} = -\infty.$$

$$(ix) \lim_{y \rightarrow 4} \frac{4-y}{2-\sqrt{y}} = \lim_{y \rightarrow 4} \frac{(4-y)(2+\sqrt{y})}{(2-\sqrt{y})(2+\sqrt{y})} = \lim_{y \rightarrow 4} \frac{(4-y)(2+\sqrt{y})}{4-y} = \lim_{y \rightarrow 4} (2+\sqrt{y}) = 4.$$

$$(x) \lim_{x \rightarrow \infty} \frac{3x+1}{2x-5} = \lim_{x \rightarrow \infty} \frac{3x}{2x} = \frac{3}{2}.$$

EXERCISE 7. Find the limits

- $\lim_{n \rightarrow \infty} a_n$,
- $\lim_{n \rightarrow \infty} b_n$,
- $\lim_{n \rightarrow \infty} a_n/b_n$, (make sure this is well-defined)

for the following sequences $(a_n)_{n \in \mathbb{N}}$, $(b_n)_{n \in \mathbb{N}}$.

- (i) $a_n = n$, $b_n = n^2$
- (ii) $a_n = 2^n$, $b_n = 3^n$
- (iii) $a_n = \frac{2n+2}{n+1}$, $b_n = \frac{n^2-1}{n^2+2n}$
- (iv) $a_n = \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right)$, $b_n = 1 + \frac{1}{n}$
*Hint: the sequence $(a_n)_{n \in \mathbb{N}}$ is called a *telescoping sum*.*

- (i)
 - $\lim_{n \rightarrow \infty} a_n = \infty$
 - $\lim_{n \rightarrow \infty} b_n = \infty$
 - $\lim_{n \rightarrow \infty} a_n/b_n = 0$
- (ii)
 - $\lim_{n \rightarrow \infty} a_n = \infty$
 - $\lim_{n \rightarrow \infty} b_n = \infty$
 - $\lim_{n \rightarrow \infty} a_n/b_n = 0$
- (iii)
 - $\lim_{n \rightarrow \infty} a_n = 2$
 - $\lim_{n \rightarrow \infty} b_n = 1$
 - $\lim_{n \rightarrow \infty} a_n/b_n = 2$
- (iv)
 - $\lim_{n \rightarrow \infty} a_n = 1$
 - $\lim_{n \rightarrow \infty} b_n = 1$
 - $\lim_{n \rightarrow \infty} a_n/b_n = 1$

EXERCISE 8. Let f and g be functions $\mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^2 - 1, & \text{for } x \leq 0, \\ -x^2, & \text{for } x > 0. \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 3x - 2, & \text{for } x \leq 2, \\ -x + 6, & \text{for } x > 2. \end{cases}$$

Draw a graph of each function. Is f continuous at $x = 0$? Is g continuous at $x = 2$?

f is not continuous at 0. g is continuous at 2.

EXERCISE 9. For what value of a is the following function $f : \mathbb{R} \rightarrow \mathbb{R}$ continuous for all x ?

$$f(x) = \begin{cases} ax - 1, & \text{for } x \leq 1, \\ 3x^2 + 1, & \text{for } x > 1. \end{cases}$$

$a = 5$